5 Study 2

Study 1 showed that both trajectory visualizations reduced VIMS relative to unmodified 360° video while costing less Presence than Dynamic FOV. However, Study 1 compared them with only one other mitigation technique and measured sickness only after each trial. Study 2 therefore compared our *Minimap* and *Overlay Path* interfaces with *Reverse Optical Flow* (ROF), a counter-optical-flow visualization. We treated Minimap and Overlay Path as the *trajectory-based* interfaces and ROF as the *non-trajectory* comparison, and asked: **are trajectory-based interfaces as effective as ROF at reducing VIMS, while being less intrusive?** Study 2 also measured sickness both before and after every trial, and collected a complete preference ranking and interview from every participant.

5.1 Participants

We recruited 36 participants. Nineteen had used a VR headset before and 17 had not. All participants completed all nine experimental conditions (three methods × three videos), yielding 324 trials and 648 SSQ administrations with no missing data.

5.2 Study Design and Procedure

Study 2 followed a 3 (*Method*: ROF, Minimap, Overlay Path) × 3 (*Video*: Walking, Cycling, Driving) within-subjects design. Unlike Study 1, it had **no Baseline condition**, so all comparisons are between mitigation techniques. Each participant experienced the three methods in separate blocks, and each block contained all three videos. Method order was fully counterbalanced: each of the six possible orders was assigned to six participants. Video order within blocks was counterbalanced so that each video appeared equally often in each position. Participants completed the SSQ immediately before each video, and the SSQ and the Presence questionnaire immediately after it. The median interval between finishing one trial and starting the next was 6.3 minutes (IQR 5.1–7.3). After all nine trials, participants ranked the three methods and took part in a short interview.

5.3 Measures

Simulator sickness. We used the same Vietnamese version of the 16-item SSQ and the same Kennedy weighted scores (TSC, N, O, D) as in Study 1, but administered the questionnaire both before and after every trial. Our main outcome was the change score $\Delta = \text{post} - \text{pre}$, where positive values mean that symptoms increased during viewing. The pre-trial scores also let us check that participants started each condition at a similar level of sickness.

Immersion. Participants completed the same five-item, 7-point presence questionnaire as in Study 1 after every trial (Cronbach's $\alpha = .78$).

Preference ranking and post-study interviews. After all nine trials, every participant ranked the three methods from 1 (best) to 3 by how well each balanced immersion and VIMS mitigation, and explained the ranking in a semi-structured interview. Later participants were also asked which method best reduced their sickness (21 named one method) and which felt most realistic (26 named one method). We coded the interview notes with a codebook of eleven codes in three families: *perceived sickness*, *perceived realism*, and *interface experience* (occlusion, distraction, glance cost, route anticipation, timing, visual noise, visibility, familiarity, and agency). Each code was counted at most once per participant and method, so all counts below are numbers of participants.

To address intrusiveness, we counted a *recorded intrusiveness-related complaint* when a participant criticized a method for blocking the scene (occlusion), drawing attention away from it (distraction), requiring gaze shifts (glance cost), or producing flicker or visual clutter (visual noise). Participants were not asked directly whether a method was intrusive, so this measure counts unprompted complaints; a method without complaints is not thereby shown to be unobtrusive.

5.4 Analysis

For the SSQ, we conducted 2 (*Time*: pre, post) × 3 (*Method*) × 3 (*Video*) repeated-measures ANOVAs. The *Time* × *Method* interaction is the effect of interest, because it tests whether the methods differed in how much sickness increased during viewing. As in Study 1, we applied Greenhouse–Geisser corrections, Holm corrections across the four SSQ outcomes, and report η^2_G. Presence was analyzed with a 3 × 3 repeated-measures ANOVA.

To compare the trajectory-based interfaces with ROF, we computed for each participant the difference between their mean ΔTSC under Minimap and Overlay Path and their ΔTSC under ROF (contrast weights $-1, \frac{1}{2}, \frac{1}{2}$). Because the question is whether the trajectory-based interfaces are *as effective as* ROF, we used a **non-inferiority test** (one-sided $\alpha = .05$) with a margin of $+5$ TSC points: the trajectory-based interfaces are non-inferior if the upper limit of the 90% confidence interval of the difference lies below $+5$. The margin is one third of the smallest reduction relative to Baseline in Study 1 (14.9 TSC points, Overlay Path), so a non-inferior method keeps at least two-thirds of that benefit. We checked the result with a mixed-effects model on all 324 trials, a participant bootstrap, and separate tests for each trajectory-based interface.

For intrusiveness, we computed for each participant the difference between whether they complained about ROF (0 or 1) and the mean of the same indicator for Minimap and Overlay Path. We tested whether this difference was positive with an exact permutation test that enumerates every assignment of the three method labels within each participant. Answering the research question in the affirmative required both the non-inferiority test and the intrusiveness test to be significant; because both must succeed, no further α adjustment is needed.

For the ranking, we used a Friedman test with Kendall's W, and a Plackett–Luce model in which Minimap and Overlay Path share one preference parameter.

The SSQ change scores were zero-inflated and heavy-tailed. Rank-transformed ANOVAs, Friedman tests, bootstrap confidence intervals, and a mixed model excluding extreme values all led to the same conclusions, so we report the parametric results below.

5.5 Results

5.5.1 VIMS and SSQ

Table 6 summarizes the ANOVA results and Table 7 the scores for each method. **Viewing increased sickness under every method**: *Time* was significant for all four SSQ outcomes, and TSC rose from 9.55 before to 17.98 after viewing (+8.44, 95% CI [5.27, 11.61], $d_z = 0.90$). **The method did not affect how much sickness increased**: the *Time* × *Method* interaction was not significant for any SSQ outcome (all $F \leq 0.78$, uncorrected $p \geq .44$, $\eta^2_G \leq .001$). As Figure 4 shows, TSC increased by 9.00 points with ROF, 9.07 with Minimap, and 7.24 with Overlay Path. Participants started each method at a similar level (pre-trial TSC 9.90, 9.38, and 9.35; all

$p \geq .77$), so these differences reflect what happened during viewing.

The **video type** mattered more than the method. The *Time* × *Video* interaction was significant for TSC, Oculomotor discomfort, and Disorientation: the Walking video increased TSC by 13.09 points, roughly 2.4 times the increase for Driving (5.44) and nearly twice that for Cycling (6.79). Walking differed from both other videos (Bonferroni $p \leq .003$), whereas Cycling and Driving did not differ. The three-way interaction was not significant, so no method was better suited to a particular video.

Symptoms partly carried over between trials: the post-trial TSC of one trial predicted the pre-trial TSC of the next ($b = 0.39$, $p < .001$). Because method order was fully counterbalanced, this carryover adds noise but does not favor any method.

5.5.2 Are Trajectory-Based Interfaces as Effective as ROF?

Table 8 and Figure 5 show the non-inferiority results. **The trajectory-based interfaces were non-inferior to ROF in measured VIMS.** On average, TSC increased 0.85 points *less* with the trajectory-based interfaces than with ROF (90% CI [−2.54, 0.84]). The upper limit was well below the +5 margin ($p < .001$), and the result held in the mixed-effects model (upper limit 2.41) and the bootstrap (0.76). **Both interfaces were individually non-inferior**: the upper limit was 2.71 for Minimap and 0.69 for Overlay Path (Holm-corrected $p \leq .002$). The difference also lay within ±5 points in a two-sided equivalence test, so the trajectory-based interfaces were not meaningfully *better* than ROF either. No SSQ subscale differed between the groups (all Holm-corrected $p \geq .83$), and the contrast did not vary by video ($F(1.73, 60.71) = 0.82$, $p = .43$).

Because Study 2 has no Baseline, this result shows that the trajectory-based interfaces were about as effective as ROF; it does not show that any of the three methods reduced VIMS relative to unmodified 360° video.

Comparison with Study 1. Applying the same contrast to Study 1, with Dynamic FOV as the non-trajectory comparison, did not establish non-inferiority: post-trial TSC was 7.8 points higher with the trajectory-based interfaces than with Dynamic FOV (95% CI [−0.9, 16.5]). This interval includes both zero and the margin, so the comparison is inconclusive. In Study 1, the trajectory-based interfaces still reduced TSC by 15.5 points relative to Baseline (95% CI [6.5, 24.4]) and produced higher Presence than Dynamic FOV (+0.55, 95% CI [0.26, 0.84]).

5.5.3 Presence

Unlike in Study 1, **Presence did not differ between methods** ($F(1.84, 64.48) = 0.25$, $p = .758$); the effects of Video and Method × Video were also not significant. The trajectory-based interfaces scored 0.02 points lower than ROF (95% CI [−0.23, 0.19]), and all three methods averaged about 5.1 on the 7-point scale. Presence was unrelated to the increase in sickness ($r = -.04$, $p = .82$). None of the three methods therefore showed the Presence cost that Dynamic FOV showed in Study 1.

5.5.4 Intrusiveness-Related Complaints

The trajectory-based interfaces did not receive fewer recorded complaints than ROF (Figure 6). Thirteen participants (36%) complained about ROF, 7 (19%) about Minimap, and 13 (36%) about Overlay Path. The mean within-participant difference was +0.08 (bootstrap 95% CI [−0.08, 0.26]; exact one-sided $p = .247$), and Minimap alone did not differ significantly from ROF (exact McNemar test, Holm-corrected $p = .36$). Instead, the complaints differed in *kind*:

ROF. Flicker and visual clutter were ROF's main problem (9 participants, compared with 1 for Minimap and none for Overlay Path), e.g., "too many motion vectors" or white streaks that caused glare. Opinions on occlusion were split (7 for, 7 against). At the same time, ROF was the method most often praised for realism (13), for example for creating a "wind effect" or blending with the surrounding scene.

Minimap. No participant said that the Minimap blocked the scene, and 13 praised it for keeping the view clear; 10 valued its preview of the route. Attention was the main issue: 5 participants found it distracting, while 5 others said they did not notice it or forgot to look at it.

Overlay Path. Anticipating upcoming turns was the most frequently cited advantage of any method in the study (17 participants), and 11 found the Overlay Path realistic. However, it also received the most distraction complaints (8), and 6 participants said that it was too large or blocked the scene.

The comparison depended on which complaint types were counted. Excluding distraction, where Overlay Path received the most complaints, produced a significant difference ($p = .012$), whereas excluding visual noise, ROF's main problem, removed any difference ($p = .70$). The interviews were coded by one researcher and inter-rater reliability has not been assessed, so we treat this analysis as exploratory.

5.5.5 Preference Ranking

Unlike in Study 1, all 36 participants provided a complete ranking (Table 9). **No method was preferred overall**: the rankings did not differ (Friedman $\chi^2(2) = 0.72$, $p = .697$), and participants barely agreed with one another (Kendall's $W = .010$). ROF was ranked last by 16 participants (44%, 95% CI [28, 62]), which was not more often than expected by chance ($p = .11$). In the Plackett–Luce model, each trajectory-based interface had an estimated probability of .36 of being ranked first, compared with .27 for ROF, but this difference was not significant (worth ratio 1.34, 95% CI [0.78, 2.32], $p = .28$).

The secondary questions showed the same picture. Of the 21 participants who named the method that best reduced their sickness, 8 chose ROF, 7 Minimap, and 6 Overlay Path; of the 26 who named the most realistic method, 11 chose ROF, 7 Minimap, and 8 Overlay Path.

When explaining their first choice, 86% of participants referred to interface experience, such as occlusion, distraction, or route anticipation, whereas only 19% referred to sickness and 25% to realism. **Participants' sense of "balance" therefore rested mainly on how each interface fit into the viewing experience**, rather than on perceived sickness reduction.

5.5.6 Study 2 Summary

Table 10 combines the quantitative and qualitative results. Study 2 shows that **the trajectory-based interfaces were non-inferior to ROF in measured VIMS**: sickness increased by a similar amount under all three methods, and the data rule out the trajectory-based interfaces being more than 0.84 TSC points worse than ROF on average. **Presence was equally high for all three methods.** The

Table 6: Repeated-measures ANOVA results for Study 2. SSQ outcomes were analyzed with a 2 (Time) $\times$ 3 (Method) $\times$ 3 (Video) ANOVA; Presence, measured after viewing only, with a 3 (Method) $\times$ 3 (Video) ANOVA. F values are reported for significant effects; Greenhouse–Geisser corrections were applied, and p values for the SSQ outcomes are Holm-corrected across the four outcomes. Time: $df = 1, 35$.

Effect	TSC	Nausea	Oculomotor	Disorientation	Presence
Time	$F = 29.15$, $p < .001$	$F = 11.08$, $p = .002$	$F = 23.97$, $p < .001$	$F = 28.64$, $p < .001$	–
Method	$p = 1.000$	$p = 1.000$	$p = 1.000$	$p = 1.000$	$p = .758$
Video	$p = .101$	$p = .253$	$F = 5.72$, $p = .021$	$p = .192$	$p = .311$
Time $\times$ Method	$p = 1.000$	$p = 1.000$	$p = 1.000$	$p = 1.000$	–
Time $\times$ Video	$F = 7.73$, $p = .004$	$p = .060$	$F = 9.61$, $p = .001$	$F = 6.91$, $p = .008$	–
Method $\times$ Video	$p = 1.000$	$p = 1.000$	$p = 1.000$	$p = 1.000$	$p = .568$
Time $\times$ Method $\times$ Video	$p = 1.000$	$p = 1.000$	$p = 1.000$	$p = 1.000$	–

Table 7: Main quantitative results of Study 2 by method, collapsed across video type (participant means $\pm$ SD, $n = 36$). Δ is the post-minus-pre change; positive values mean symptoms increased during viewing.

Method	Pre TSC	Post TSC	ΔTSC	ΔNausea	ΔOculomotor	ΔDisorientation	Presence
ROF	9.90 ± 18.65	18.91 ± 21.18	9.00 ± 9.56	3.97 ± 6.24	7.65 ± 9.86	13.66 ± 15.45	5.10 ± 1.04
Minimap	9.38 ± 15.02	18.46 ± 20.51	9.07 ± 11.39	3.27 ± 9.33	8.91 ± 11.83	12.63 ± 16.32	5.12 ± 1.05
Overlay Path	9.35 ± 19.45	16.59 ± 21.59	7.24 ± 12.60	2.56 ± 7.93	6.67 ± 12.28	10.96 ± 17.22	5.04 ± 1.06

Table 8: Non-inferiority of the trajectory-based interfaces. Difference in TSC points (trajectory minus comparison; positive = more sickness). Non-inferior when the upper limit of the 90% CI is below the $+5$ margin. Study 2 uses the pre-to-post change; Study 1 uses post-trial TSC.

Comparison	Difference	90% CI	Non-inferior
Study 2 (vs ROF)			
Trajectory (pooled)	-0.85	$[-2.54, 0.84]$	**yes**
Minimap	$+0.07$	$[-2.57, 2.71]$	**yes**
Overlay Path	-1.77	$[-4.22, 0.69]$	**yes**
Study 1 (vs Dynamic FOV)			
Trajectory (pooled)	$+7.81$	[0.54, 15.08]	no
Minimap	$+7.20$	[0.16, 14.25]	no
Overlay Path	$+8.42$	$[-0.08, 16.91]$	no

Table 9: Preference ranking in Study 2 ($n = 36$). Lower mean rank indicates stronger preference. Borda score: 2 points for first place, 1 for second.

Method	Mean rank	Borda	First	Last
Overlay Path	**1.92**	**39**	**13**	**10**
Minimap	1.97	37	11	**10**
ROF	2.11	32	12	16

data did *not* show that the trajectory-based interfaces were less intrusive: they drew about as many recorded complaints as ROF, but of a different kind. ROF was criticized for flicker, Overlay Path for distraction and occlusion, and Minimap drew the fewest complaints. Participants did not agree on a single most balanced method; their rankings followed these trade-offs rather than perceived sickness reduction.

These results have three limitations. First, Study 2 had no Baseline, so it cannot show that any method reduced VIMS relative to unmodified video. Second, sickness was mild (post-trial TSC 16.6–18.9, compared with 48.3–71.6 in Study 1), which may have left little room for the methods to differ. Third, the complaint analysis relies on unprompted comments coded by one researcher.

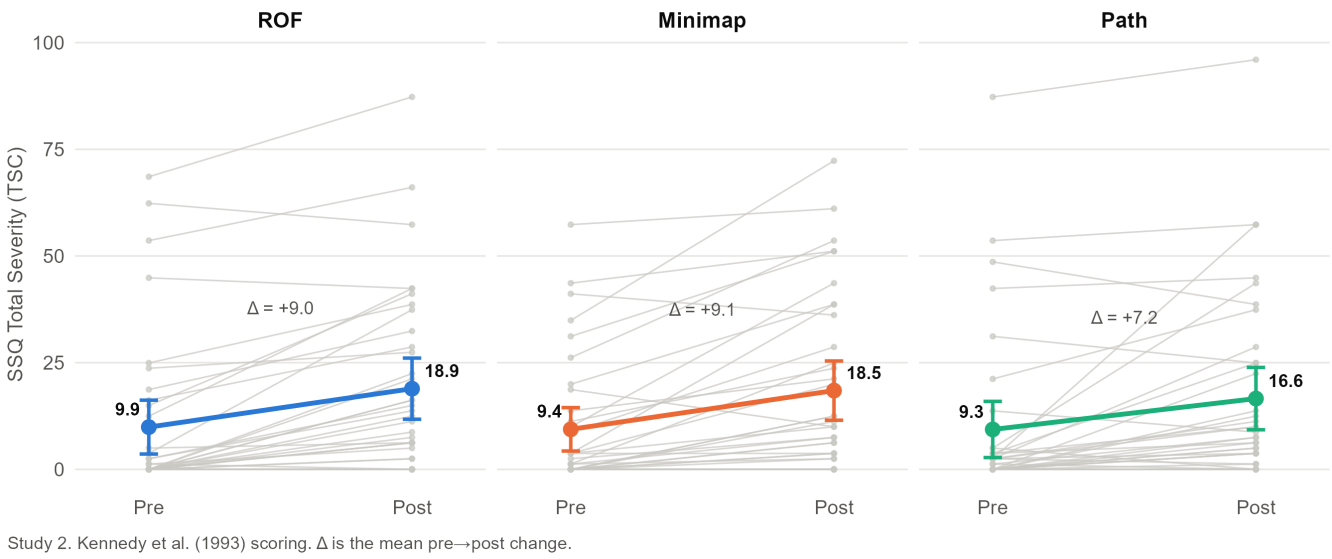


Figure 4: SSQ Total Severity before and after each video in Study 2. Grey lines are individual participants (means over the three videos); colored points show the mean with 95% CI. Sickness rose by a similar amount under all three methods.

Table 10: Integration of quantitative and qualitative findings in Study 2. Interview counts are numbers of participants (out of 36).

Method	Quantitative result	Interview feedback
Minimap	TSC increased by 9.07; non-inferior to ROF (upper limit 2.71). Presence 5.12.	Fewest intrusiveness-related complaints (7). Never described as blocking the scene; route preview valued (10). Some found it distracting (5) or forgot to look at it (5).
Overlay Path	Numerically smallest TSC increase (7.24); non-inferior to ROF (upper limit 0.69). Presence 5.04.	Most-cited advantage in the study: anticipating turns (17); seen as realistic (11). Most distraction complaints (8); too large or blocking the scene (6).
ROF	TSC increased by 9.00. Presence 5.10.	Most often praised for realism (13), e.g., a "wind effect". Most complaints about flicker and visual clutter (9). Ranked last most often (16).

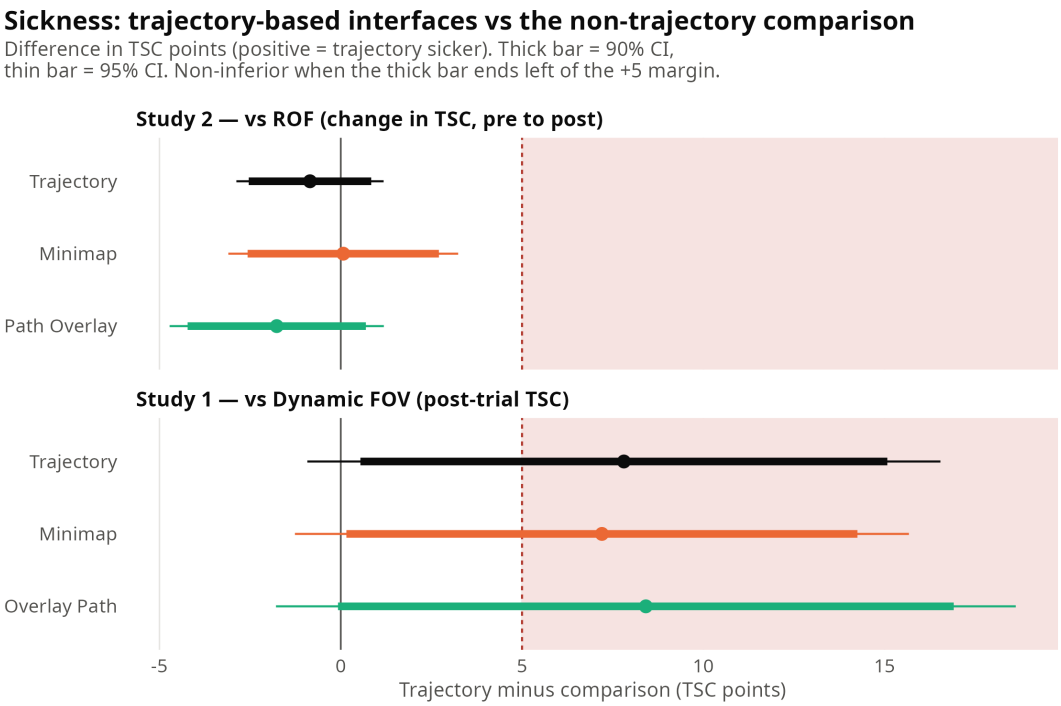


Figure 5: Sickness difference between the trajectory-based interfaces and the non-trajectory comparison in each study. Thick bars show the 90% CI and thin bars the 95% CI; the dashed line marks the +5 TSC non-inferiority margin.

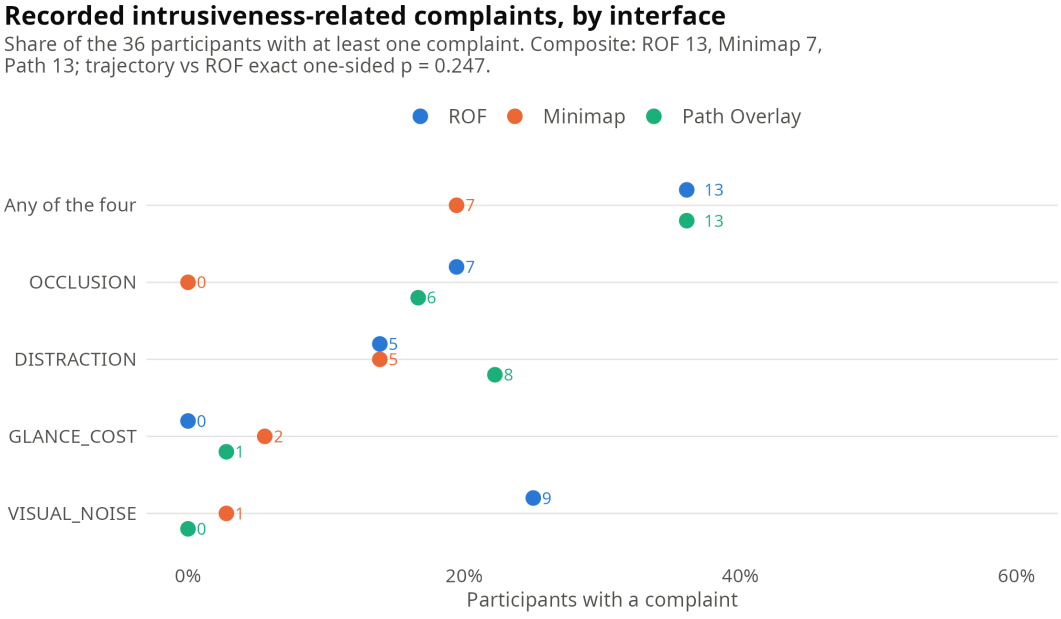


Figure 6: Participants who made at least one recorded intrusiveness-related complaint about each method, overall ("any of the four") and by complaint type. Counts are out of 36 participants.